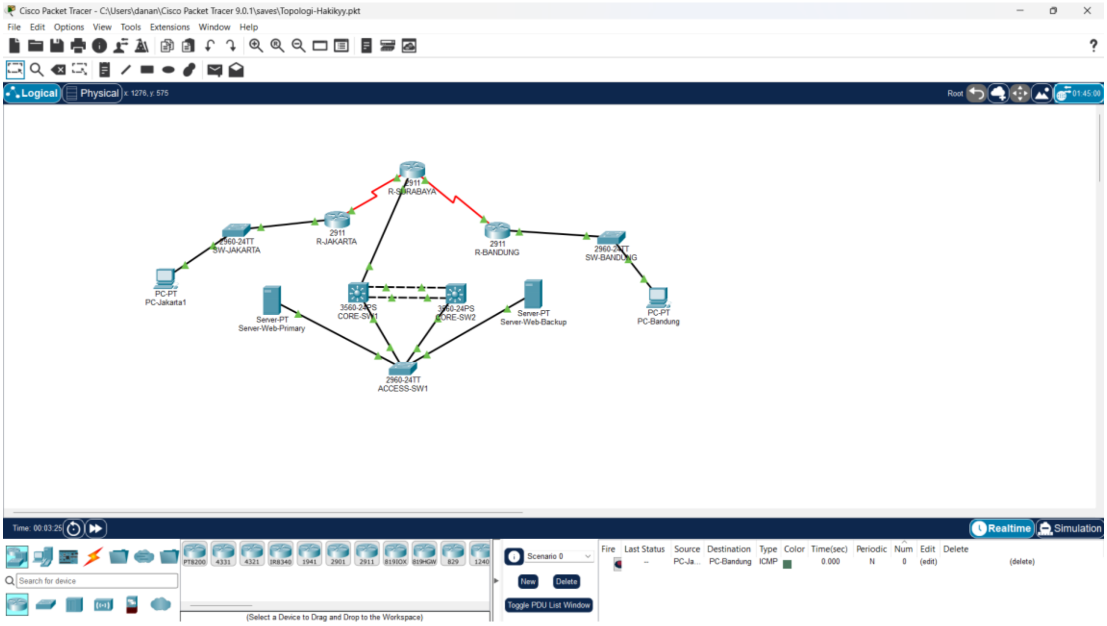
DevOps Engineer Intern – Tahapan Seleksi MagangHub | SEVIMA

Nama : Mohammad Khusai Hakiki

1. SOAL NOMOR 1 - Infrastructure Provisioning
   1. Topologi Keseluruhan



   2. Bukti Konfigurasi per Device (CLI)

Device	Command	Hasil
CORE-SW1	show etherchannel summary	1 Po1(SU) LACP Fa0/1(P) Fa0/2(P) CORE-SW1>show standby brief ^ % Invalid input detected at '^' marker. CORE-SW1>show standby brief ^ % Invalid input detected at '^' marker. CORE-SW1>show standby brief ^ % Invalid input detected at '^' marker. CORE-SW1>clear Translating "clear"...domain server (255.255.255.255)
CORE-SW1	show standby brief	CORE-SW1#show standby brief P indicates configured to preempt. | Interface Grp Pri P State Active Standby Virtual IP Vl10 1 110 P Active local 10.10.10.3 10.10.10.254 CORE-SW1#
CORE-SW2	show standby brief	CORE-SW2>enable CORE-SW2#show standby brief P indicates configured to preempt. | Interface Grp Pri P State Active Standby Virtual IP Vl10 1 90 Standby 10.10.10.2 local 10.10.10.254 CORE-SW2#
CORE-SW1 & CORE-SW2	show vlan brief	CORE-SW1#show standby brief P indicates configured to preempt. | Interface Grp Pri P State Active Standby Virtual IP Vl10 1 110 P Active local 10.10.10.3 10.10.10.254 CORE-SW1#show vlan brief VLAN Name Status Ports ---- -------------------------------- --------- ------------------------------- 1 default active Po1, Fa0/1, Fa0/2, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2 10 SERVER-FARM active 1002 fddi-default active 1003 token-ring-default active 1004 fddinet-default active 1005 trnet-default active CORE-SW2#show vlan brief VLAN Name Status Ports ---- -------------------------------- --------- ------------------------------- 1 default active Po1, Fa0/1, Fa0/2, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2 10 SERVER-FARM active 1002 fddi-default active 1003 token-ring-default active 1004 fddinet-default active 1005 trnet-default active

<table>
<tr><td>ACCESS-SW1</td><td>show interfaces trunk</td><td><pre>
ACCESS-SW1#show interfaces trunk
Port   Mode   Encapsulation Status   Native vlan
Fa0/1   on   802.1q   trunking   1
Fa0/2   on   802.1q   trunking   1

Port   Vlans allowed on trunk
Fa0/1   1-1005
Fa0/2   1-1005

Port   Vlans allowed and active in management domain
Fa0/1   1,10
Fa0/2   1,10

Port   Vlans in spanning tree forwarding state and not pruned
Fa0/1   10
Fa0/2   1,10
</pre></td></tr>
<tr><td>R-SURABAYA</td><td>show ip interface brief</td><td><pre>
R-SURABAYA#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   10.10.1.2   YES manual up   up
GigabitEthernet0/1   unassigned   YES NVRAM administratively down down
GigabitEthernet0/2   unassigned   YES NVRAM administratively down down
Serial0/0/0   192.168.100.1   YES manual up   up
Serial0/0/1   192.168.100.5   YES manual up   up
Vlan1   unassigned   YES unset administratively down down
</pre></td></tr>
<tr><td>R-SURABAYA</td><td>show ip ospf neighbor</td><td><pre>
R-SURABAYA#show ip ospf neighbor

Neighbor ID   Pri   State   Dead Time   Address   Interface
10.10.10.2   1   FULL/BDR   00:00:38   10.10.1.1   GigabitEthernet0/0
192.168.100.2   0   FULL/ -   00:00:34   192.168.100.2   Serial0/0/0
192.168.100.6   0   FULL/ -   00:00:34   192.168.100.6   Serial0/0/1
</pre></td></tr>
<tr><td>R-SURABAYA</td><td>show ip route</td><td><pre>
R-SURABAYA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C   10.10.1.0/30 is directly connected, GigabitEthernet0/0
L   10.10.1.2/32 is directly connected, GigabitEthernet0/0
O   10.10.10.0/24 [110/2] via 10.10.1.1, 00:16:19, GigabitEthernet0/0
O   10.20.10.0/24 [110/65] via 192.168.100.2, 00:17:24, Serial0/0/0
O   10.30.10.0/24 [110/65] via 192.168.100.6, 00:17:24, Serial0/0/1
     192.168.100.0/24 is variably subnetted, 4 subnets, 2 masks
C   192.168.100.0/30 is directly connected, Serial0/0/0
L   192.168.100.1/32 is directly connected, Serial0/0/0
C   192.168.100.4/30 is directly connected, Serial0/0/1
L   192.168.100.5/32 is directly connected, Serial0/0/1
</pre></td></tr>
<tr><td>R-JAKARTA</td><td>show ip route</td><td><pre>
R-JAKARTA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
O   10.10.1.0/30 [110/65] via 192.168.100.1, 00:17:41, Serial0/0/0
O   10.10.10.0/24 [110/66] via 192.168.100.1, 00:17:21, Serial0/0/0
C   10.20.10.0/24 is directly connected, GigabitEthernet0/0
L   10.20.10.1/32 is directly connected, GigabitEthernet0/0
O   10.30.10.0/24 [110/129] via 192.168.100.1, 00:18:11, Serial0/0/0
     192.168.100.0/24 is variably subnetted, 3 subnets, 2 masks
C   192.168.100.0/30 is directly connected, Serial0/0/0
L   192.168.100.2/32 is directly connected, Serial0/0/0
O   192.168.100.4/30 [110/128] via 192.168.100.1, 00:18:11, Serial0/0/0
</pre></td></tr>
<tr><td>R-BANDUNG</td><td>show ip route</td><td><pre>
R-BANDUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
O   10.10.1.0/30 [110/65] via 192.168.100.5, 00:18:10, Serial0/0/0
O   10.10.10.0/24 [110/66] via 192.168.100.5, 00:17:45, Serial0/0/0
O   10.20.10.0/24 [110/129] via 192.168.100.5, 00:18:40, Serial0/0/0
C   10.30.10.0/24 is directly connected, GigabitEthernet0/0
L   10.30.10.1/32 is directly connected, GigabitEthernet0/0
     192.168.100.0/24 is variably subnetted, 3 subnets, 2 masks
O   192.168.100.0/30 [110/128] via 192.168.100.5, 00:18:50, Serial0/0/0
C   192.168.100.4/30 is directly connected, Serial0/0/0
L   192.168.100.6/32 is directly connected, Serial0/0/0
</pre></td></tr>
</table>

3. Bukti Konektivitas End-to-End

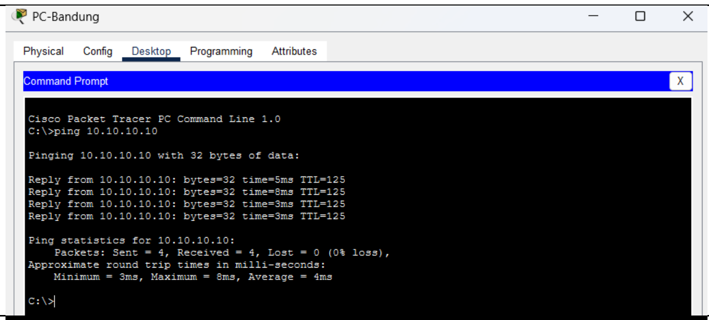
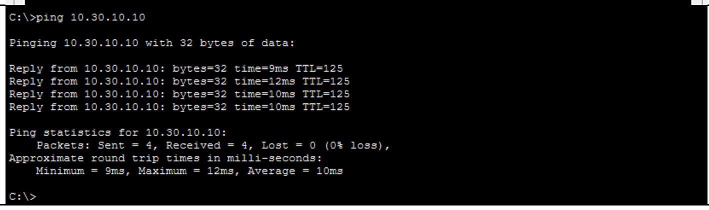
<table>
<tr><td>PC-Jakarta1 Ke Server-Web-Primary</td><td>PC-Jakarta1
Physical Config Desktop Programming Attributes
Command Prompt X<pre>
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time=6ms TTL=125
Reply from 10.10.10.10: bytes=32 time=3ms TTL=125
Reply from 10.10.10.10: bytes=32 time=4ms TTL=125
Reply from 10.10.10.10: bytes=32 time=5ms TTL=125

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 4ms

C:\>
</pre></td></tr>
</table>

PC-Bandung Ke Server-Web-Backup	
PC-Jakarta1 Ke PC-Bandung	

2. SOAL NOMOR 2: Make Your Web Great Again

A. Task A. Linux Hardening

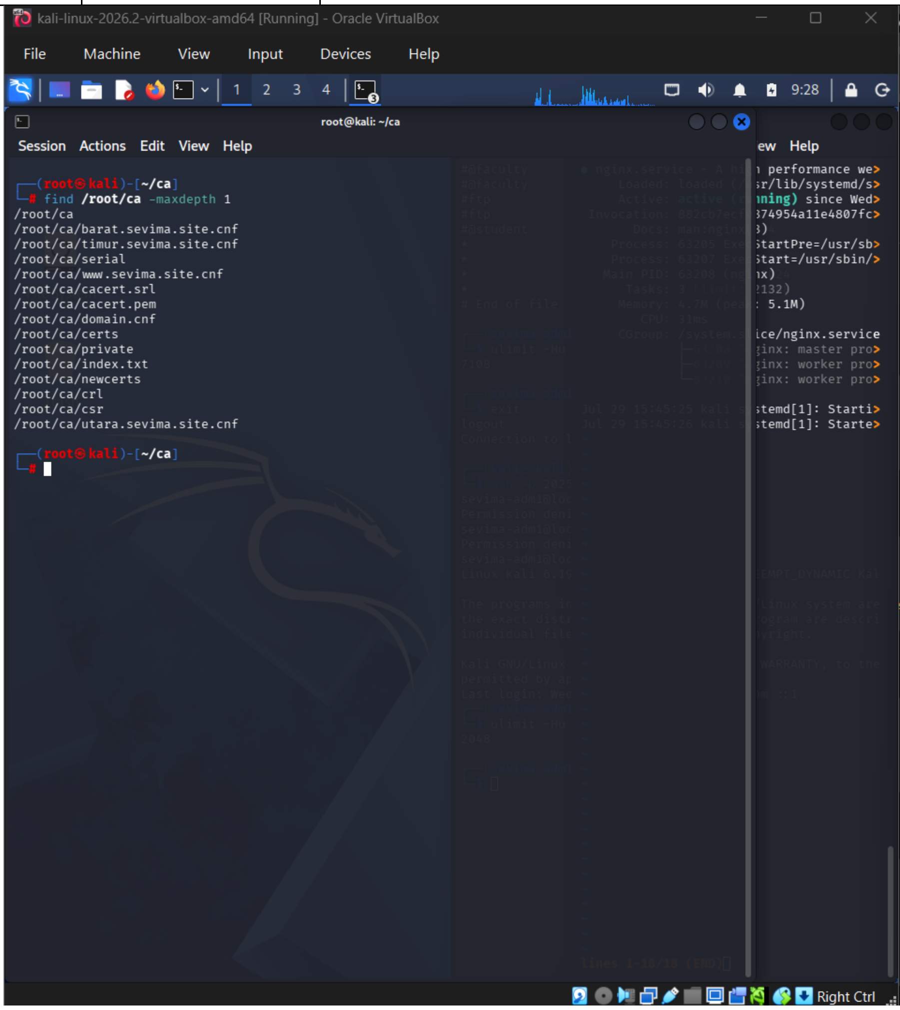
No	Langkah	Command
1	Melakukan identifikasi sistem operasi	cat /etc/os-release
2	Mengetahui user aktif	whoami
3	Mengetahui konfigurasi IP	ip a
4	Menginstall paket keamanan	apt update && apt install rsyslog auditd acct apache2 nginx haproxy openssl socat -y
5	Mengaktifkan service logging	systemctl enable --now rsyslog
6	Mengaktifkan audit daemon	systemctl enable --now auditd
7	Mengaktifkan process accounting	systemctl enable --now acct
8	Verifikasi service	systemctl status rsyslog auditd acct
9	Membuat user administrator	useradd -m -s /bin/bash sevima-adm1
10	Mengatur password user	passwd sevima-adm1
11	Mengatur limit resource	nano /etc/security/limits.conf
12	Verifikasi PAM Limits	grep pam_limits /etc/pam.d/common-session
13	Login sebagai user	su - sevima-adm1
14	Verifikasi limit	ulimit -a

B. Task B. Certificate Authority (CA)

- B.1 Membuat Struktur CA

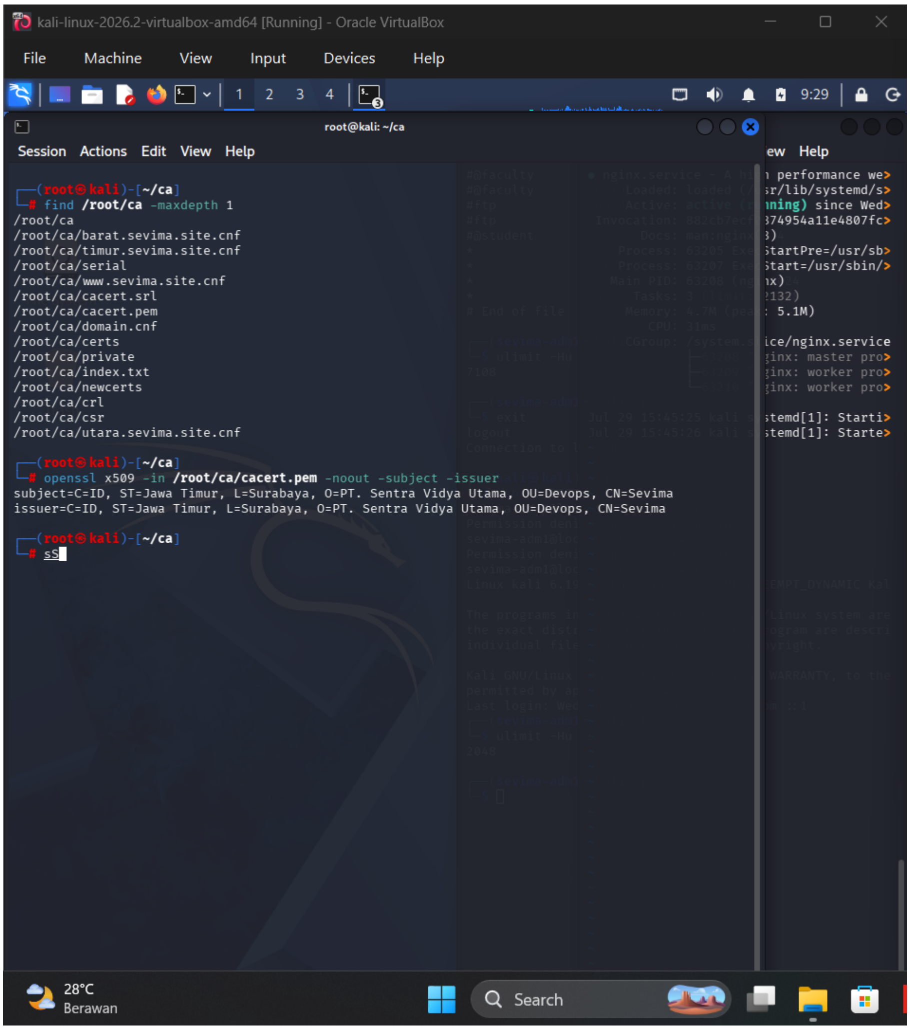
No	Langkah	Command
1	Membuat direktori CA	mkdir -p /root/ca/{certs,private,newcerts,crl,csr}
2	Membuat database CA	touch /root/ca/index.txt
3	Membuat serial number	echo 1000 > /root/ca/serial

4	Verifikasi direktori	find /root/ca -maxdepth 1



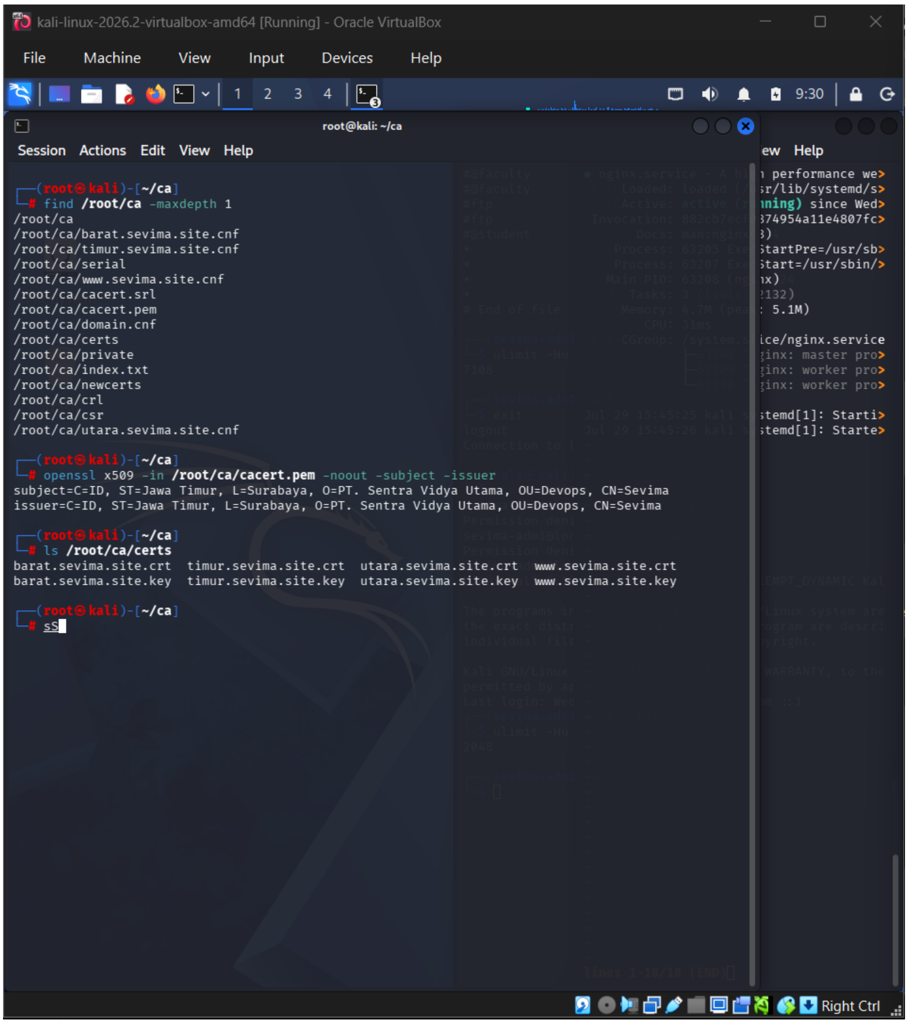
- B.2 Membuat Root CA

No	Langkah	Command
1	Generate Root Private Key	openssl genrsa -aes256 -out private/cacert.key 4096
2	Generate Root Certificate	openssl req -x509 -new -key private/cacert.key -sha256 -days 3650 -out cacert.pem
3	Verifikasi Root CA	openssl x509 -in cacert.pem -noout -subject -issuer



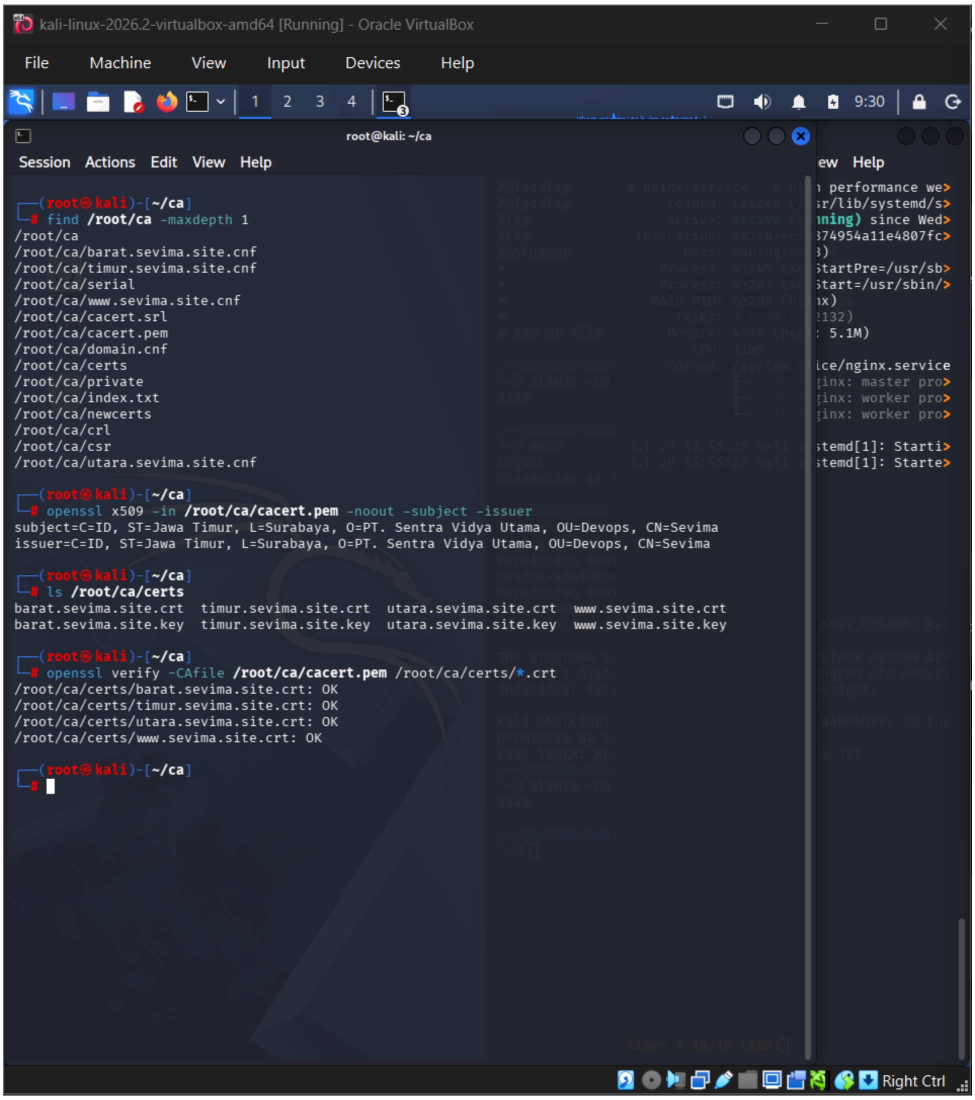
- B.3 Membuat Sertifikat Domain

No	Langkah	Command
1	Membuat file konfigurasi CSR	nano /root/ca/domain.cnf
2	Generate Private Key seluruh domain	openssl genrsa
3	Generate CSR	openssl req -new
4	Sign Certificate menggunakan Root CA	openssl x509 -req ...
5	Melakukan seluruh proses otomatis	for DOMAIN in ... do ... done



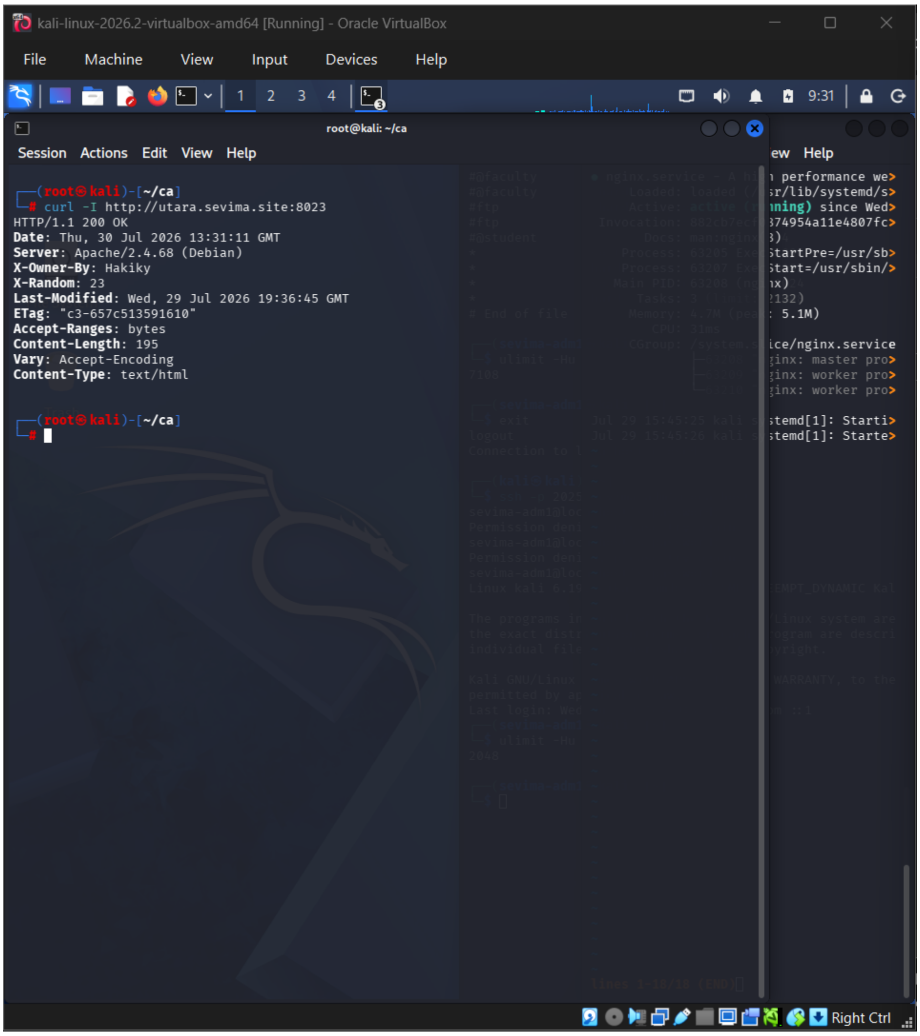
- B.4 Verifikasi Certificate

No	Langkah	Command
1	Melihat hasil sertifikat	ls /root/ca/certs
2	Verifikasi Certificate	openssl verify -CAfile /root/ca/cacert.pem /root/ca/certs/*.crt



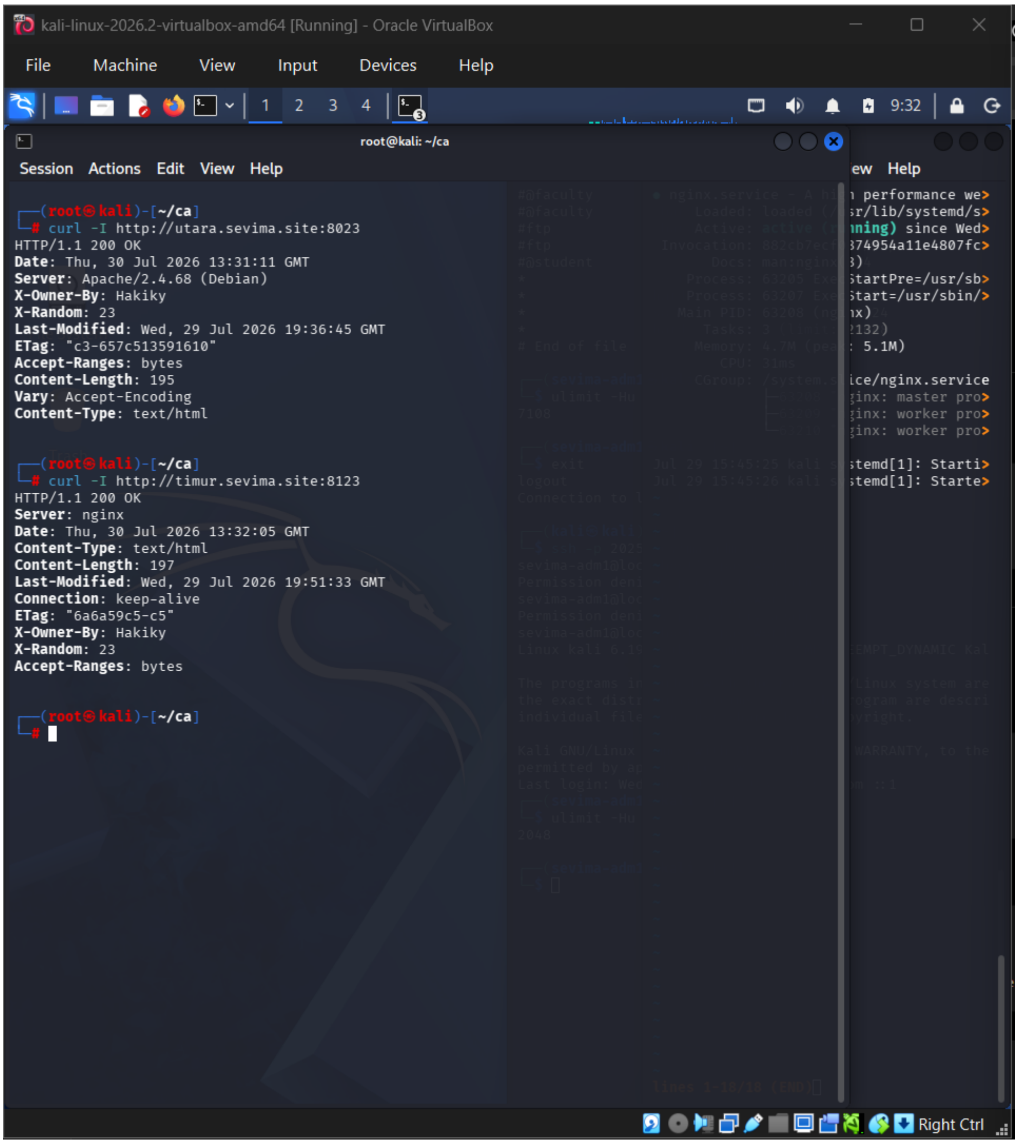
C. Task C. Apache Web Server

No	Langkah	Command
1	Menonaktifkan VirtualHost default	a2dissite 000-default.conf
2	Mengaktifkan module headers	a2enmod headers
3	Membuat Document Root	mkdir -p /var/www/utara
4	Membuat halaman web	nano /var/www/utara/index.html
5	Menambahkan port Apache	nano /etc/apache2/ports.conf
6	Membuat VirtualHost	nano /etc/apache2/sites-available/utara.conf
7	Mengaktifkan VirtualHost	a2ensite utara.conf
8	Restart Apache	systemctl restart apache2
9	Menambahkan hosts	nano /etc/hosts
10	Pengujian	curl -I http://utara.sevima.site:8023



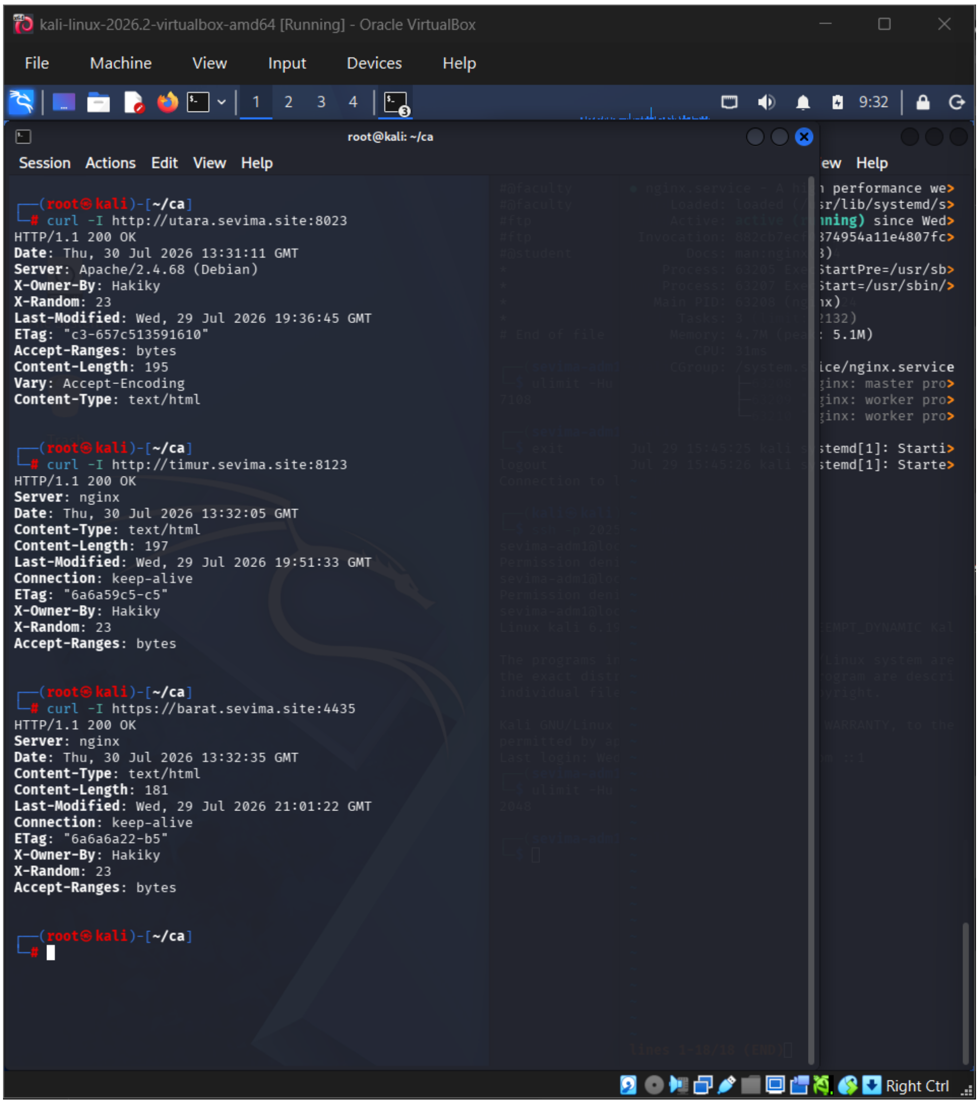
C. Task C. Nginx Web Server

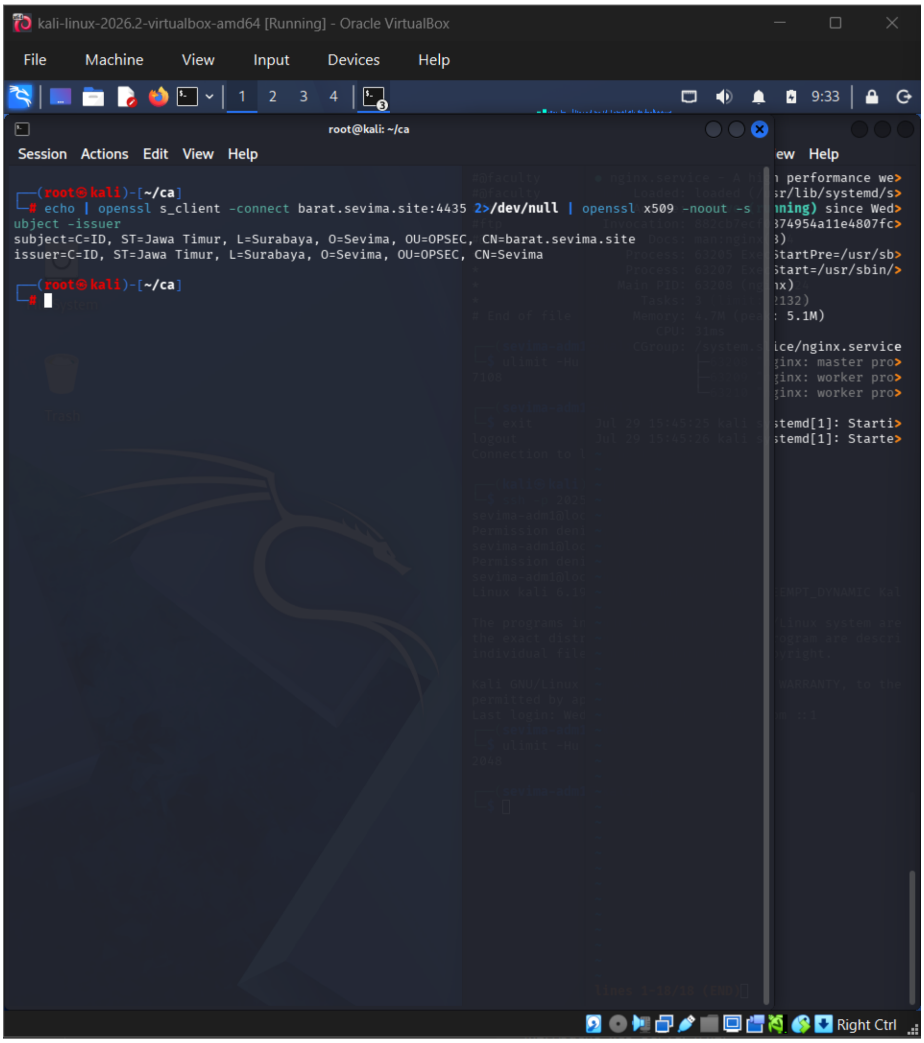
No	Langkah	Command
1	Membuat Document Root	mkdir -p /var/www/timur
2	Membuat halaman web	nano /var/www/timur/index.html
3	Membuat VirtualHost	nano /etc/nginx/sites-available/timur
4	Mengaktifkan site	ln -s /etc/nginx/sites-available/timur /etc/nginx/sites-enabled/
5	Verifikasi konfigurasi	nginx -t
6	Restart Nginx	systemctl restart nginx
7	Menambahkan hosts	nano /etc/hosts
8	Pengujian	curl -I http://timur.sevima.site:8123


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
9:32
root@kali: ~/ca
Session Actions Edit View Help
┌──(root㉿kali)-[~/ca]
└─# curl -I http://utara.sevima.site:8023
HTTP/1.1 200 OK
Date: Thu, 30 Jul 2026 13:31:11 GMT
Server: Apache/2.4.68 (Debian)
X-Owner-By: Hakiky
X-Random: 23
Last-Modified: Wed, 29 Jul 2026 19:36:45 GMT
ETag: "c3-657c513591610"
Accept-Ranges: bytes
Content-Length: 195
Vary: Accept-Encoding
Content-Type: text/html
┌──(root㉿kali)-[~/ca]
└─# curl -I http://timur.sevima.site:8123
HTTP/1.1 200 OK
Server: nginx
Date: Thu, 30 Jul 2026 13:32:05 GMT
Content-Type: text/html
Content-Length: 197
Last-Modified: Wed, 29 Jul 2026 19:51:33 GMT
Connection: keep-alive
ETag: "6a6a59c5-c5"
X-Owner-By: Hakiky
X-Random: 23
Accept-Ranges: bytes
┌──(root㉿kali)-[~/ca]
└─#
Right Ctrl

C. Task C. HTTPS Server

No	Langkah	Command
1	Membuat Document Root	mkdir -p /var/www/barat
2	Membuat halaman web	nano /var/www/barat/index.html
3	Membuat HTTPS VirtualHost	nano /etc/nginx/sites-available/barat
4	Mengaktifkan site	ln -s /etc/nginx/sites-available/barat /etc/nginx/sites-enabled/
5	Verifikasi konfigurasi	nginx -t
6	Restart Nginx	systemctl restart nginx
7	Pengujian HTTPS	curl -I https://barat.sevima.site:4435
8	Verifikasi Certificate	openssl s_client -connect barat.sevima.site:4435 -showcerts

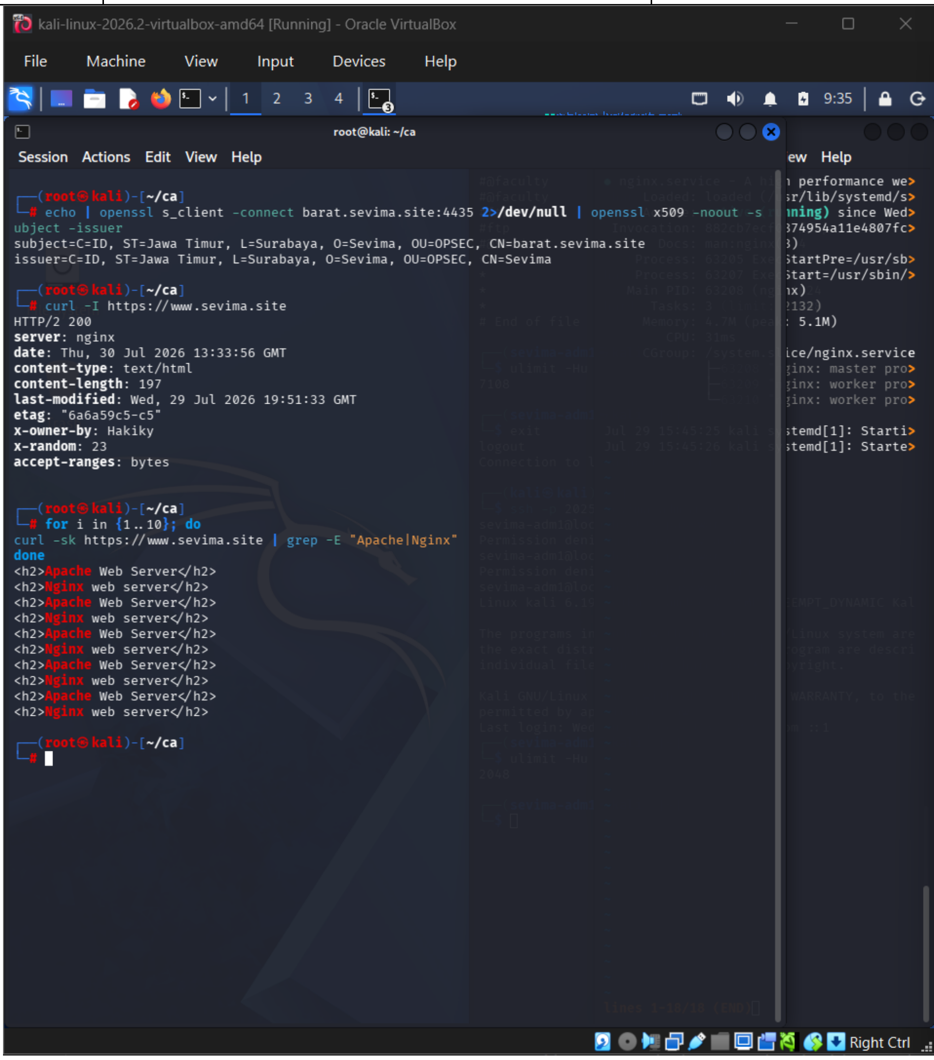




D. Task D. HAProxy Load Balancer

No	Langkah	Command
1	Install HAProxy	apt install haproxy -y
2	Membuat file PEM	cat www.sevima.site.crt www.sevima.site.key > /etc/haproxy/certs/www.sevima.site.pem
3	Backup konfigurasi	cp /etc/haproxy/haproxy.cfg /etc/haproxy/haproxy.cfg.bak
4	Mengubah konfigurasi HAProxy	nano /etc/haproxy/haproxy.cfg
5	Validasi konfigurasi	haproxy -c -V -f /etc/haproxy/haproxy.cfg
6	Restart HAProxy	systemctl restart haproxy
7	Enable HAProxy	systemctl enable haproxy
8	Pengujian HTTP	curl -I http://www.sevima.site
9	Pengujian HTTPS	curl -I https://www.sevima.site

10	Pengujian Round Robin	for i in {1..10}; do curl -sk https://www.sevima.site | grep -E "Apache|Nginx"; done



E. Pengujian Akhir

No	Tujuan	Command
1	Verifikasi Root CA	openssl x509 -in cacert.pem -noout -subject -issuer
2	Verifikasi seluruh Certificate	openssl verify -CAfile cacert.pem certs/*.crt
3	Apache	curl -I http://utara.sevima.site:8023
4	Nginx	curl -I http://timur.sevima.site:8123
5	HTTPS	curl -I https://barat.sevima.site:4435

6	HAProxy HTTPS	curl -I https://www.sevima.site
7	Round Robin	for i in {1..10}; do curl -sk https://www.sevima.site | grep -E "Apache|Nginx"; done